each sphere will be deflected by the same, but smaller than before angle θ from the vertical. ($\theta < 3.0°$) **B1**

5 **(a)** Acceleration is directly proportional to the displacement. **B1**

Acceleration is always in the opposite direction to its displacement **B1**

(b) **(i)** 0.034 m **B1**

(ii)

$$\omega = \frac{v_0}{x_0}$$

$$= \frac{0.60}{0.034}$$ **C1**

$$= 17.647$$

$$= 18 \text{ rad s}^{-1}$$ **A1**

(c) **(i)** Upwards is taken to be positive from the graph,

$$F_{net} = N - W = -m\omega^2 x$$ **C1**

When the object loses contact, $N = 0$

$$mg = m\omega^2 x$$

$$x = \frac{g}{\omega^2}$$

$$= \frac{9.81}{17.647^2}$$ **C1**

$$= 0.0315$$

$$= 0.032 \text{ m}$$ **A1**

(ii) C marked at (0.032, 0.2) **B1**